

V3 Pipeline — Egitim Raporu

Tarih: 2026-03-22 22:01

PIPELINE OZELLIKLERI:

Feature Engineering: V3 (Label-encoded predictions + ayri OHE + manuel k-mer)
Feature sayisi: 979
Toplam ornek: 4287 (CV: 2138, Hold-Out: 858)
Model turleri: LightGBM, XGBoost
Hyperparameter: Grid Search (12 combo/model, 3-fold CV)
Degerlendirme: 5-Fold Stratified CV + %20 Hold-Out Test

5-FOLD CV SONUCLARI (egitim seti %80):

lightgbm : F1 = 0.9745 +/- 0.0049
xgboost : F1 = 0.9763 +/- 0.0045

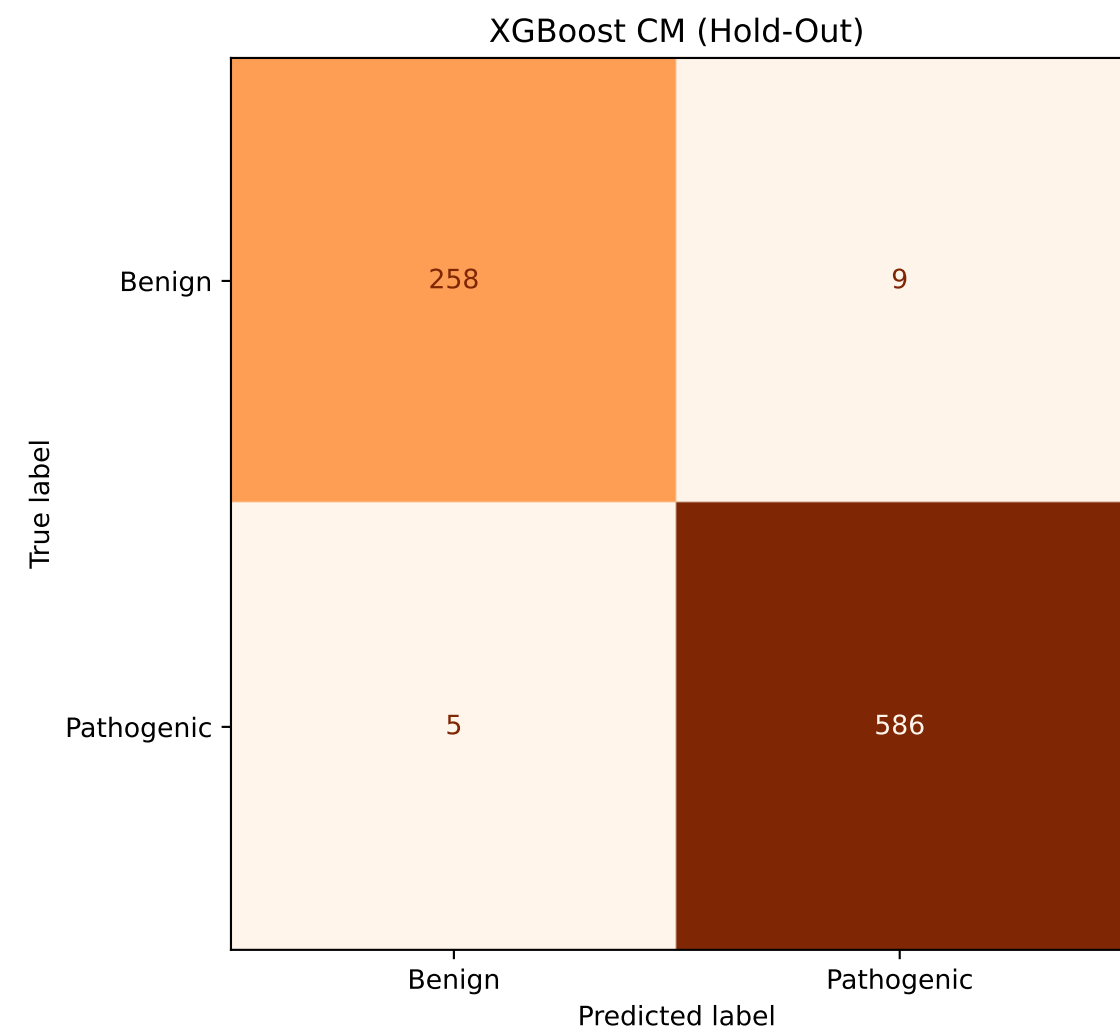
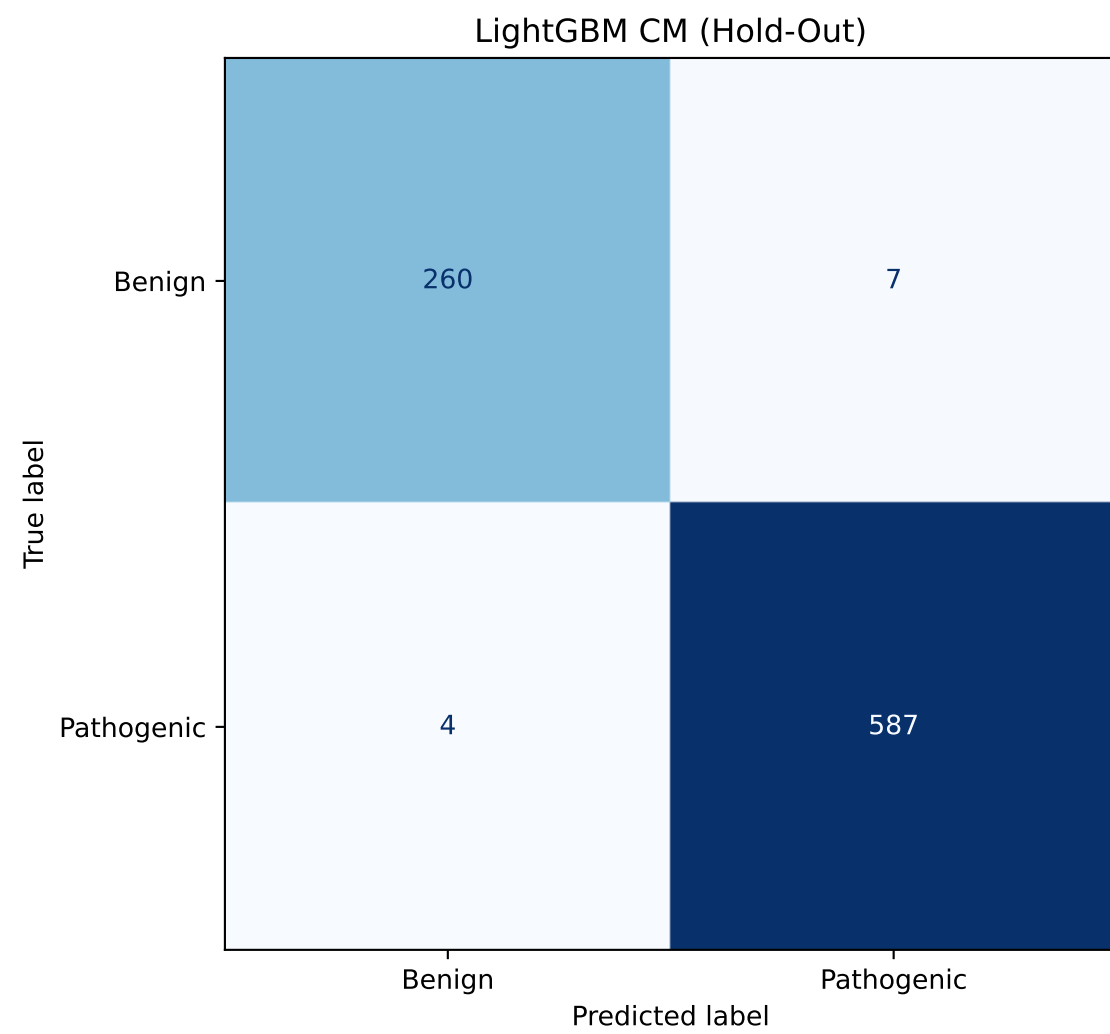
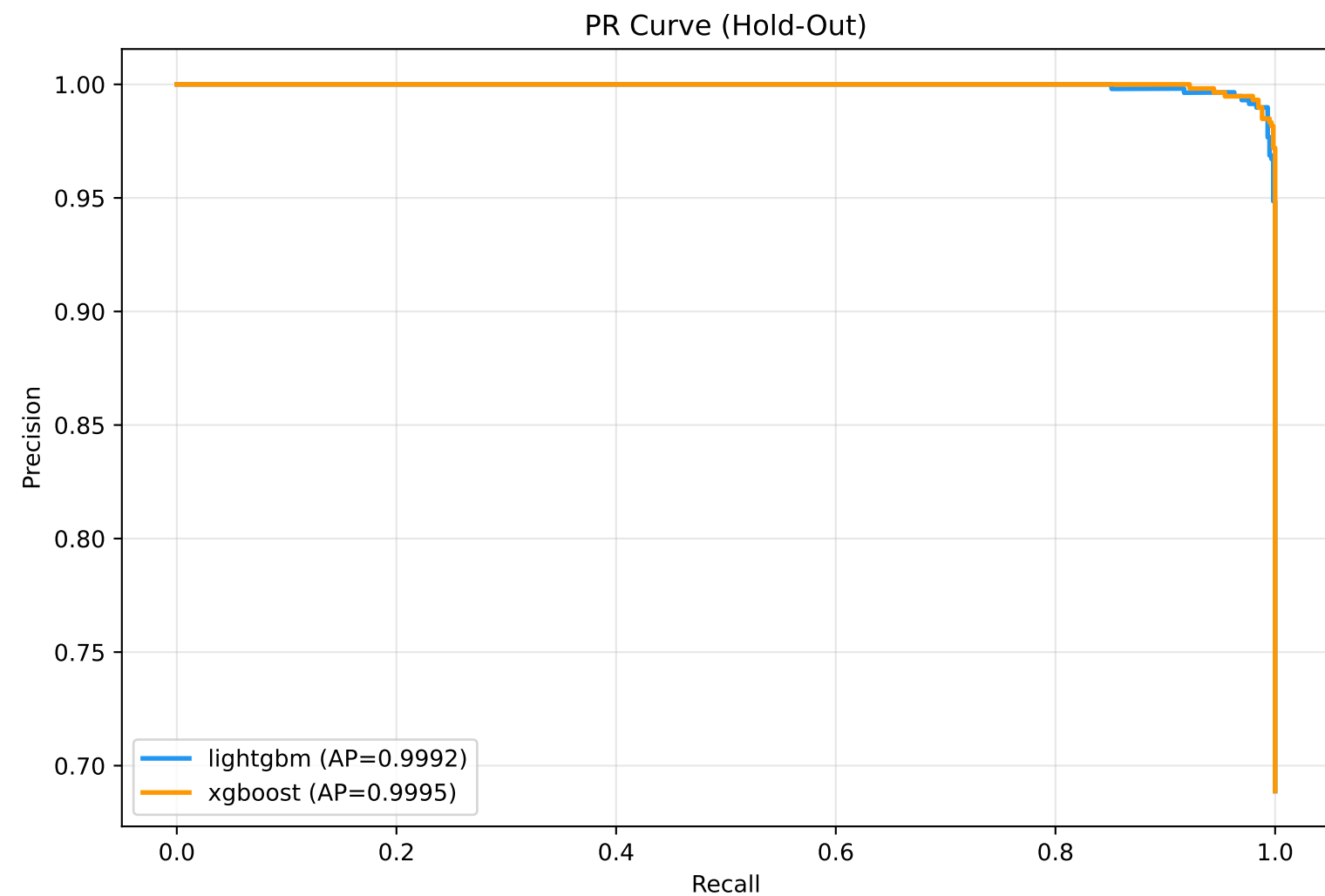
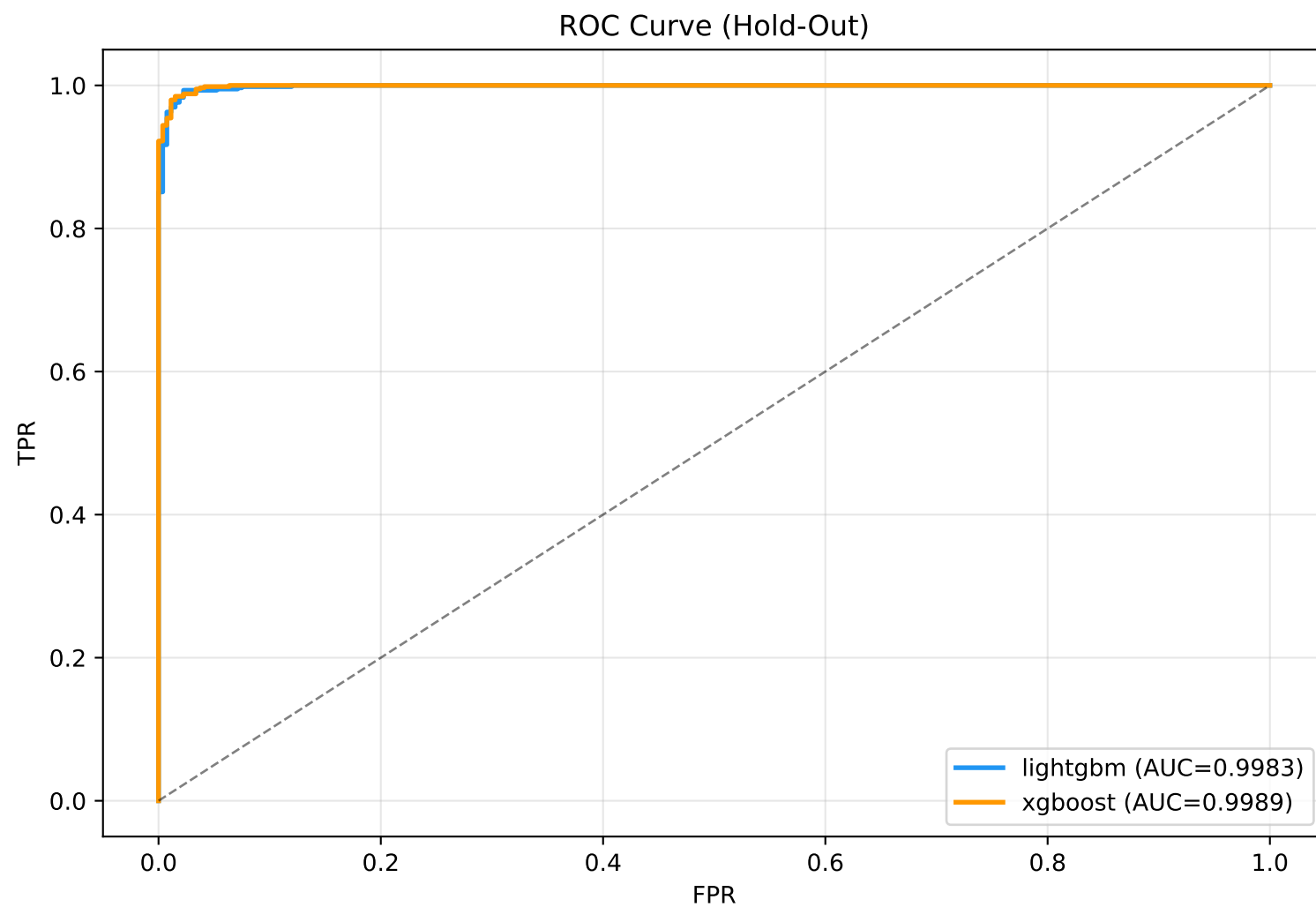
HOLD-OUT TEST SONUCLARI (hic gorulmemis %20):

lightgbm : F1=0.9907 AUC=0.9983 (CV F1=0.9745)
xgboost : F1=0.9882 AUC=0.9989 (CV F1=0.9763)

KARSILASTIRMA (full_fe vs v3 hold-out):

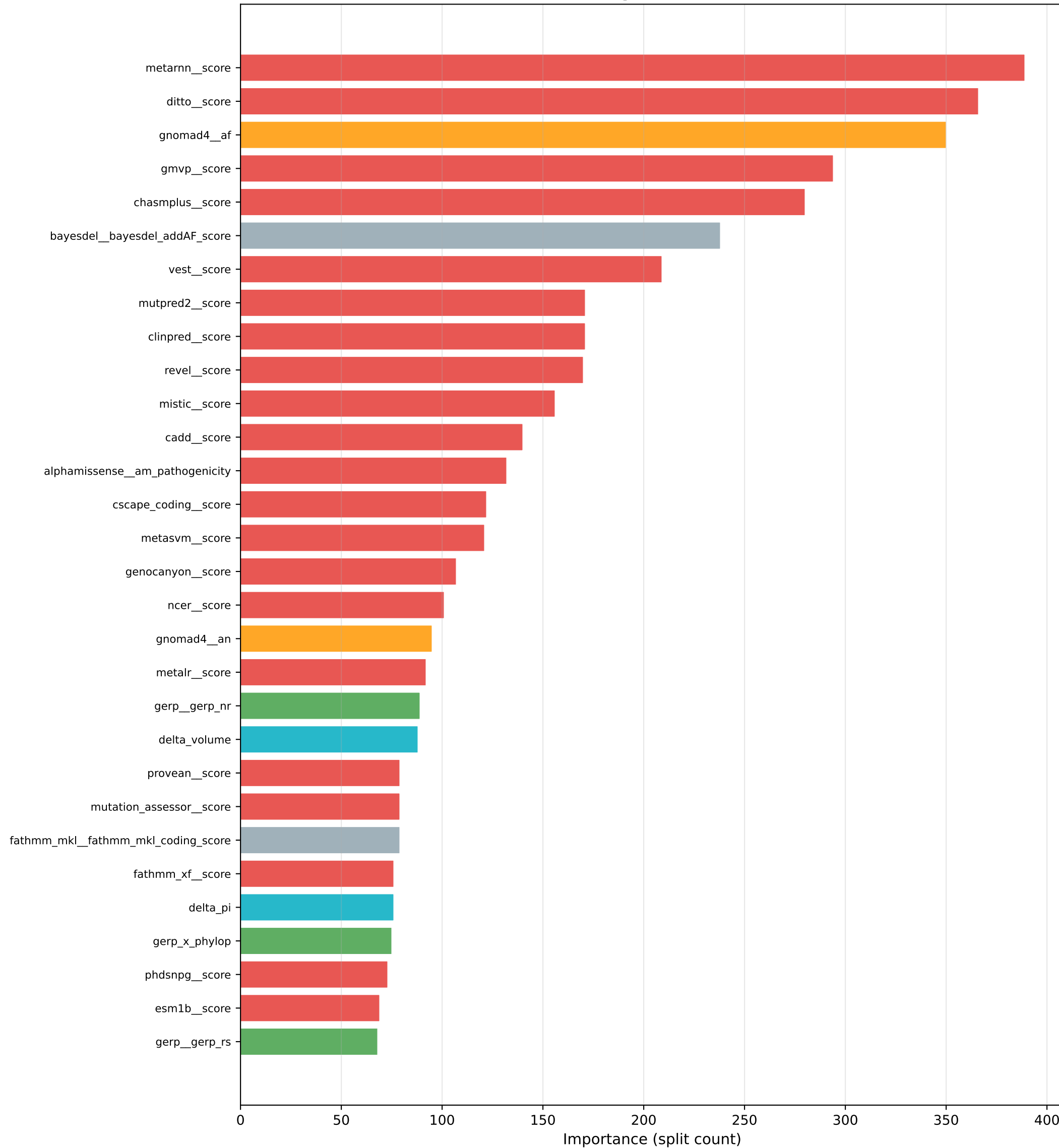
lightgbm: full_fe F1=0.6967 -> v3 Hold-Out F1=0.9907 (+0.2941)
xgboost: full_fe F1=0.6879 -> v3 Hold-Out F1=0.9882 (+0.3003)

Hold-Out Test Sonuclari (%20 — Hic Gorulmemis Veri)

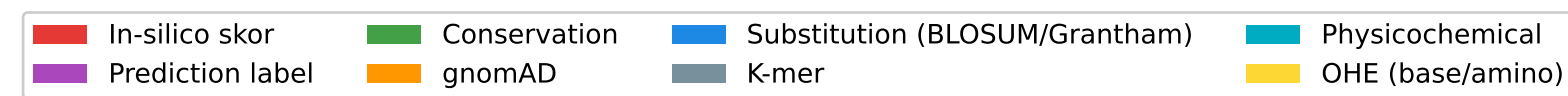
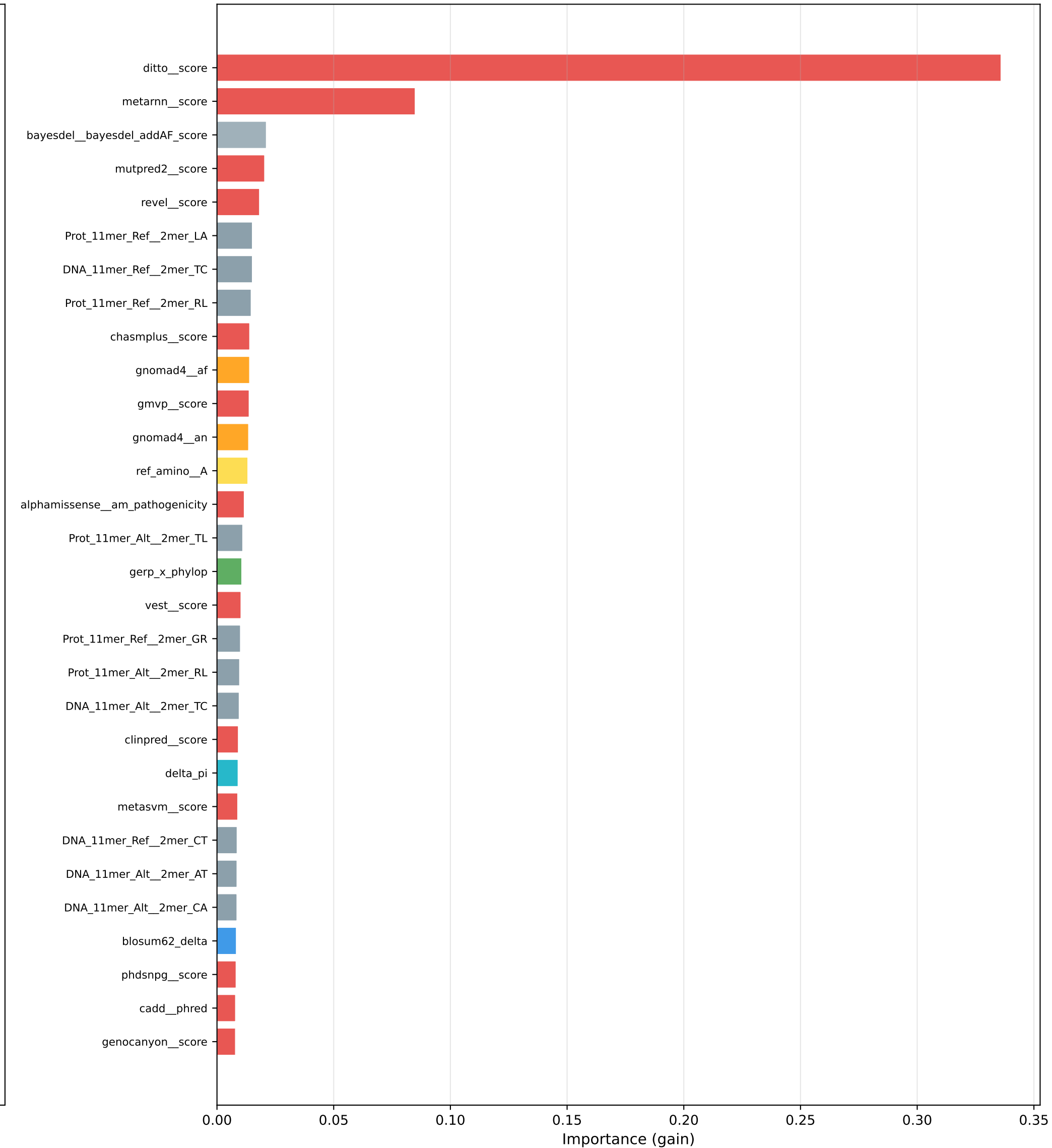


Feature Importance — Top 30 (En İyi CV Fold Modeli)

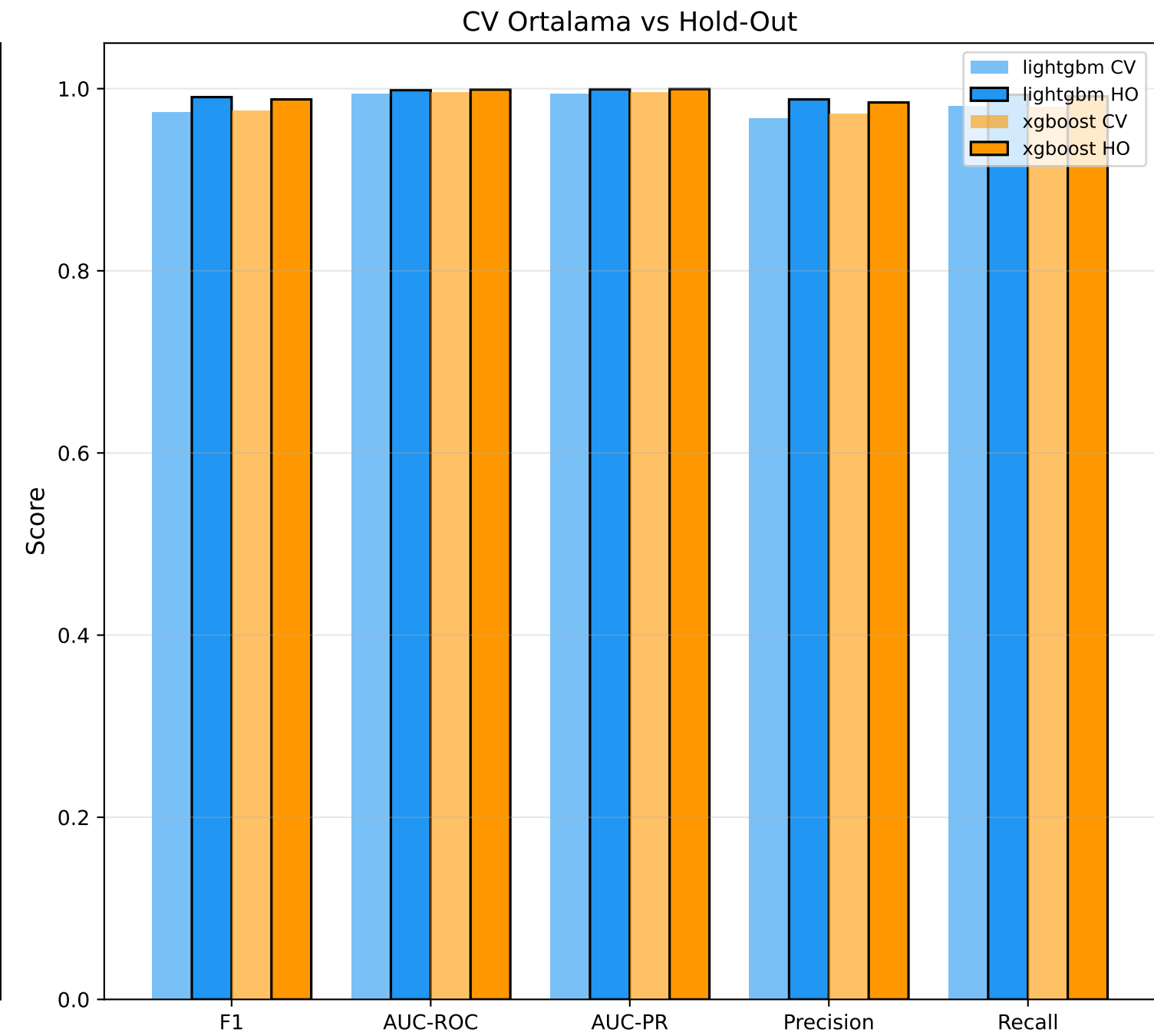
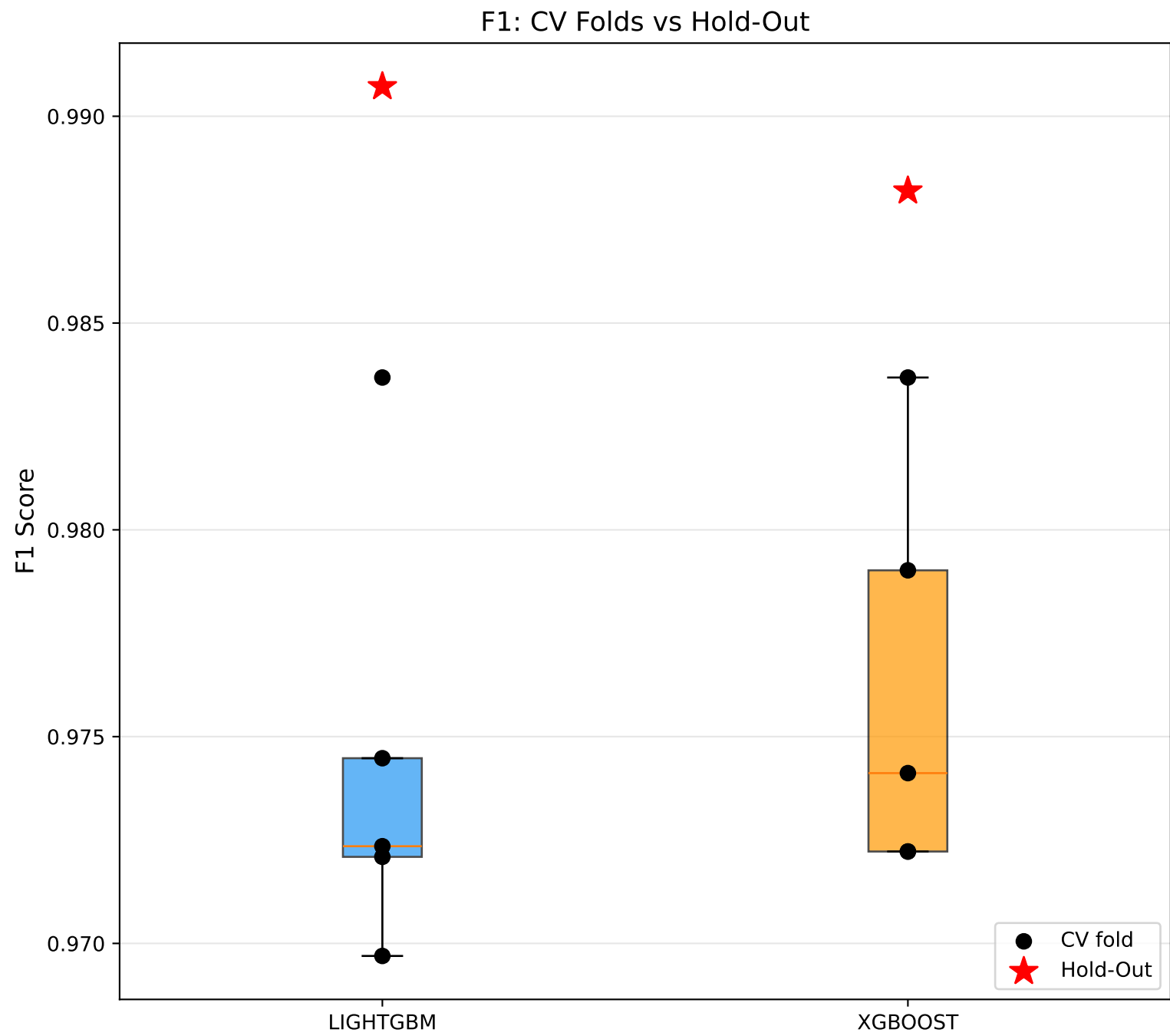
LightGBM



XGBoost



CV vs Hold-Out Karsilastirma



Detayli Sonuclar: 5-Fold CV + Hold-Out Test

Fold	Model	Split	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
1	lightgbm	CV	0.9745	0.9973	0.9974	0.9677	0.9813	0.9487
2	lightgbm	CV	0.9721	0.9909	0.9865	0.9676	0.9766	0.9440
3	lightgbm	CV	0.9724	0.9957	0.9957	0.9591	0.9860	0.9443
4	lightgbm	CV	0.9837	0.9951	0.9967	0.9814	0.9860	0.9672
5	lightgbm	CV	0.9697	0.9951	0.9958	0.9630	0.9765	0.9392
1	xgboost	CV	0.9722	0.9969	0.9970	0.9633	0.9813	0.9441
2	xgboost	CV	0.9790	0.9944	0.9944	0.9767	0.9813	0.9580
3	xgboost	CV	0.9741	0.9958	0.9956	0.9810	0.9673	0.9487
4	xgboost	CV	0.9837	0.9971	0.9976	0.9814	0.9860	0.9672
5	xgboost	CV	0.9722	0.9956	0.9961	0.9589	0.9859	0.9442
ORT	lightgbm	CV	0.9745	0.9948	0.9944	0.9678	0.9813	0.9487
ORT	xgboost	CV	0.9763	0.9960	0.9961	0.9723	0.9804	0.9524
-	lightgbm	HOLD-OUT	0.9907	0.9983	0.9992	0.9882	0.9932	0.9700
-	xgboost	HOLD-OUT	0.9882	0.9989	0.9995	0.9849	0.9915	0.9618